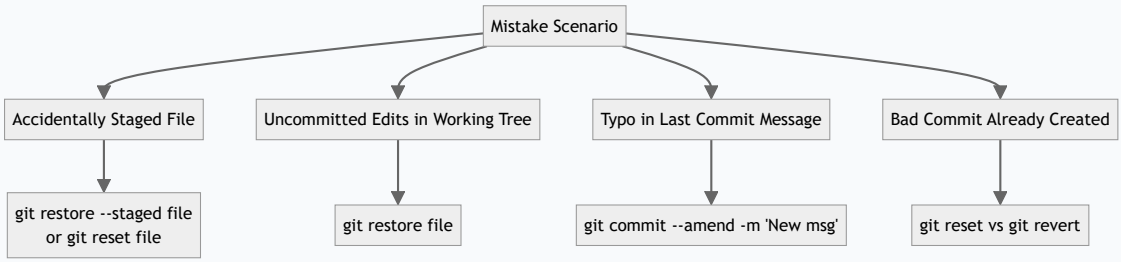
Submodule 35.19: Fixing Mistakes (Undo, Reset, Revert)

1. Common Developer Errors in Git

Mistakes are an unavoidable aspect of software engineering. Git provides powerful mechanisms to undo errors at every stage of the lifecycle:



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2. Fixing Staging & Working Tree Mistakes

1. Unstaging an Accidental File (Staged *o* Working Tree):

If you accidentally ran `git add secret_password.txt`:

```
# Modern command
git restore --staged secret_password.txt

# Classic command
git reset HEAD secret_password.txt
```

(The file remains on disk with your edits intact, but transitions back to the red untracked/modified state).

2. Discarding Uncommitted Edits in Working Tree:

If you modified a file and want to discard all changes, restoring it to the last commit's state:

```
# Modern command
git restore app.js
```

```
# Classic command
git checkout -- app.js
```

> [!WARNING] Discarding working tree edits is permanent and cannot be undone.

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3. Fixing Commit Mistakes

1. Amending the Last Commit:

If you committed and forgot to include a file, or made a typo in the commit message:

```
# Fix message only
git commit --amend -m "Corrected commit message"

# Add forgotten file to last commit
git add forgotten_file.py
git commit --amend --no-edit
```

2. Undoing Commits: `git revert` vs `git reset`

Command	Mechanism	History Impact	Production Safety
`git revert <commit_hash>`	Creates a **new commit** that mathematically inverts the changes of the specified commit.	Preserves full linear history.	**100% Safe** for shared/public branches on GitHub.
`git reset HEAD~1`	Rewinds the `HEAD` pointer backwards, removing the commit from history.	Rewrites commit history.	Dangerous on public branches; use only for private local commits.

The Three Modes of `git reset`:

1. `git reset --soft HEAD~1` : Removes the commit, but keeps changes staged in the index (ready to re-commit).
2. `git reset --mixed HEAD~1` *(default)*: Removes the commit and unstages changes, keeping modifications in your working directory.
3. `git reset --hard HEAD~1` : Completely obliterates the commit and wipes out all file modifications on disk.